

Homework

Preliminary Mathematics Standard

Earning Money

1 Questions

1.1 Ways of Earning

1. Some employers offer a superannuation bonus scheme. They pay you 9.5%, then for every additional 1% of your salary that you contribute, they match it with a further 1%. A software saleswoman makes \$70 000 per year and decides to contribute 3% of her salary into superannuation.
 - (a) How much does the saleswoman contribute from her own salary into superannuation each year?
 - (b) In total, how much does she receive into her superannuation fund each year?
$$\text{Total rate} = 9.5\% + 3\%_{(\text{employee})} + 3\%_{(\text{employer match})} = 15.5\%$$
2. There is a proposal to have the same penalty rates for both Saturday and Sunday of time-and-a-half. Currently workers receive time-and-a-half for Saturdays and double-time for Sundays. A waiter's hourly rate is \$17.75.
 - (a) By what percentage will she be worse off if the penalty rates change?
 - (b) By what percentage will her hourly rate need to increase so she earns the same amount working both days under the new rates?
Let the new hourly rate be r . We need:
3. A factory worker receives an hourly rate of \$28.40 for a standard 38-hour week. She receives time-and-a-half for any hours worked above 38.
 - (a) On average she works 42.5 hours each week. Calculate her weekly wage before tax.
 - (b) Will she be better off on a wage or a salary of \$68 000? (Assume 52 weeks.)

4. An employer proposes: removal of 17.5% annual leave loading, a 5% increase in annual wages, and 10.5% superannuation. Should employees accept? Justify with calculations.

Assume a base annual wage of W .

5. A new children's book author is offered \$100 000 cash or a 4% royalty on sales. What would sales need to be to match the cash offer?
6. A used-car salesman earns \$1 500 per month plus 6.2% commission on all sales. He sold \$243 540 worth of cars last month. What is his total wage?
7. A songwriter received \$11 473.75 in royalties for \$458 950 in sales. Find the royalty percentage.

8. A shoe manufacturer offers two pay options:

- Option 1: \$8.25 per pair of shoes made
- Option 2: 3.5% commission on shoes sold

A shoemaker makes 15 pairs/day; each pair sells for \$75.95. In one week, 250 pairs are sold. Which option earns more?

9. A telecommunications salesperson earns a base wage of \$450/week plus 2.5% commission. One week his earnings were \$502.25.
- (a) What was the total value of telephone plans sold?
- (b) If each plan was \$95, how many plans did he sell?
- (c) The salesperson is offered the choice to remain on a weekly base wage of \$530 with an increase of 3.5% in commission on all sales, or to receive an annual salary of \$37 500.
- Which option would you recommend he choose? Justify your answer using calculations.

1.2 Taxation

1. A computer technician's annual salary is \$67 374. He claims \$1 580 in deductions and receives \$225 in dividends from shares.
- (a) The technician is paid monthly. Calculate his gross monthly salary.
- (b) His employer withholds \$1 200 PAYG tax each month. Calculate the total amount of tax withheld for the year.
- (c) Determine the tax payable by the computer technician for the financial year. Hence, determine whether he receives a tax refund or is required to pay more tax, and state the amount.

2. A hairdresser receives a weekly wage of \$930.60, which includes 38 hours of normal pay plus 4 hours of overtime, paid at time-and-a-half.
 - (a) Explain how his hourly rate can be determined, and hence find his hourly rate.
 - (b) Determine his annual salary if he works an average of 40 hours a week (38 hours at normal pay and 2 hours overtime). Assume 52 weeks in 1 year.
 - (c) Using your answer from part b, determine the amount of tax payable at the end of the financial year.
 - (d) The hairdresser forgot to claim \$1 980 in deductions and is looking forward to receiving \$1 980 in a tax refund. By recalculating the amount of tax payable at the end of the financial year based on his new taxable income, explain why he won't receive \$1 980 in a tax refund.
3. Use the tax table below to answer the questions that follow.

Taxable income	Tax on this income
\$0–\$18 200	Nil
\$18 201–\$37 000	19c for each \$1 over \$18 200
\$37 001–\$87 000	\$3 572 plus 32.5c for each \$1 over \$37 000
\$87 001–\$180 000	\$19 822 plus 37c for each \$1 over \$87 000
\$180 001 and over	\$54 232 plus 45c for each \$1 over \$180 000

The Australian tax system is a tiered system, as shown in the table. A dentist's annual salary is \$97065.

- (a) Explain why the tax payable is not found by calculating 37% of \$97065.
 - (b) Explain how the additional 'plus amount' of \$19822 is calculated.
4. A mechanical engineer pays \$14812 in tax.
 - (a) Explain how her taxable income can be determined.
 - (b) Determine her taxable income.
 - (c) After the financial year she realises that she forgot to claim \$985 in tax deductions. determine her taxable income and the tax refund she should expect.
5. Each year, an IT technician moves up one level on the pay scale shown below.

Level	Annual salary
1-1	\$51 758
1-2	\$53 258
1-3	\$55 333
1-4	\$57 508
1-5	\$60 088

The technician is currently on level 1-3. At the start of the following year she moves up the pay scale to level 1-4.

- Determine the percentage change in her wage correct to 2 decimal places. A new working agreement is reached which increases the pay level of technicians on levels 1 and 2 by 3.75%.
- Determine the new percentage change in the technician's wage from her original 1-3 wage correct to 2 decimal places.
- After the pay increase of 3.75%, the technician moves to level 2-3 with an annual salary of \$65880. The overall percentage change in her wage is 8.5%. Determine the salary for level 2-2 before and after the pay increase.

2 Answers

2.1 Ways of Earning

1. (a) \$2 100
(b) \$8 750
2. (a) She will be worse off by 25% for every hour she works on Sunday, but no worse off for working on Saturdays.
(b) 16.65%
3. (a) \$1 270.90
(b) Her weekly salary would be \$1 307.69, which is higher than her hourly wage calculation (\$1 270.90). She should take the salary offer.
4. The new agreement gives 15.5% more than the base annual salary, compared to 10.8% under the current agreement.

$$\text{Proposed agreement: } \frac{57\,750 - 50\,000}{50\,000} \times 100 = 15.5\%$$

$$\text{Current agreement: } \frac{55\,423 - 50\,000}{50\,000} \times 100 = 10.8\%$$

Employees should accept the proposed agreement.

5. \$2 500 000
6. \$16 599.48
7. The royalty percentage can be found by dividing the royalty amount by the total sale amount and multiplying by 100:

$$\frac{11\,473.75}{458\,950} \times 100 = 2.5\%$$

The songwriter's royalty percentage is 2.5%.

8. Option 2 earns more money.
9. (a) \$2 090
(b) 22 plans
(c) If he sells 22 new telephone plans each week, taking the salary option of \$37 500 per annum would give him a higher weekly wage.

2.2 Taxation

1. (a) \$5 614.50
 (b) \$14 400
 (c) He will receive a refund of \$1 396.83.
2. (a) \$21.15
 Working 4 hours overtime equates to the same pay as working 6 hours normal time ($4 \times 1.5 = 6$). Calculate the number of 'normal' hours worked ($38 + 6 = 44$) and divide into the wage.
 (b) \$45 091.80
 (c) \$6 201.84
 (d) \$5 558.34
 He won't receive \$1 980 as a tax refund because the tax payable is a percentage of his taxable income. His taxable income was reduced by \$1 980 but the tax payable was reduced by 32.5% of the difference ($0.325 \times (\$8 091.80 - \$1 980)$).
3. (a) When calculating tax using this tiered system, the taxable salary of \$97 605 is split into the different tax-bracket amounts, which are found by the difference between the lower and upper limits, e.g.: $37\ 000 - 18\ 200 = \$18\ 800$.

$$\$18\ 200 + \$18\ 800 + \$50\ 000 + \$10\ 605 = \$97\ 605$$

Each amount is multiplied by a different percentage, as shown.

Tax bracket	Amount of salary (\$)	Tax %	Tax paid (\$)
\$0–\$18 200	18 200	0	0
\$18 201–\$37 000	18 800	19	$0.19 \times 18\ 800 = 3\ 572.00$
\$37 001–\$87 000	50 000	32.5	$0.325 \times 50\ 000 = 16\ 250.00$
\$87 001–\$180 000	10 605	37	$0.37 \times 10\ 605 = 3\ 923.85$
Total	\$97 605		Total = \$23 745.85

This is much lower than $37\% \times \$97\ 605 = \$36\ 113.85$.

- (b) The dentist's taxable salary falls into the fourth tax bracket, 37% for amounts over \$87 000 plus \$19 822. The \$19 822 is the tax amount from the previous tax brackets ($19\% \times \$18\ 800 + 32.5\% \times \$50\ 000$), as shown.
4. (a) Determine the tax bracket her taxable income falls within. Since she paid more than \$3572 and less than \$19 822, her taxable income is within the third tax bracket:

$$\$3572 + 32.5\% \times (\text{taxable income} - 37000)$$

(b) \$71 584.62

(c) Refund: \$320.13

5. (a) 3.93%

(b) 7.83%

(c) Before pay increase: \$60 718.89 After pay increase: \$62 996.85